Final Project
Analyzing and Predicting Smartphone Addiction from Real World Usage Data

Hunter Ryan

AI Usage Note: AI tools were used to assist with portions of the coding implementation, debugging, and code refinement, and the writing of the notation and problem formulation section throughout the project. However, the overall project design, including the selection of the four classification models, the linear regression task, evaluation methodology, the organization and structure of the LaTeX report, and the choice of visualizations and plots were independently developed and decided by me.

1 Introduction

Smartphones have become an essential part of daily life, but excessive usage has raised growing concerns about potential addictive behavior. Many people spend significant amounts of time on their devices for activities such as social media, gaming, and communication often at the expense of sleep, productivity, and overall well-being. As smartphone usage continues to increase, understanding the patterns associated with problematic use has become increasingly important.

This project examines smartphone usage data to identify factors associated with addiction and evaluate the relative importance of behavioral, demographic, and contextual features. Behavioral features such as screen time, app usage frequency, and sleep duration are analyzed to uncover patterns that distinguish between different levels of usage. Studying these patterns provides insight into how digital behavior affects people and highlights the importance of maintaining a balanced relationship with smartphones.

2 Notation, Problem Formulation, and Goals

2.1 Notation

Let $X \in \mathbb{R}^{n \times d}$ denote the dataset of smartphone usage behaviors, where n is the number of individuals and d is the number of features describing usage patterns. Each observation $x_i \in \mathbb{R}^d$ includes variables such as screen time, social media usage, app activity, sleep duration and other indicators.

2.2 Problem Formulation

The primary task is formulated as a binary classification problem where the target variable $y \in \{0, 1\}^n$ indicates whether the person is classified as addicted to smartphone usage ($y_i = 1$) or not ($y_i = 0$). The goal is to estimate a function that maps behavioral features to an addiction prediction.

The secondary task is formulated as a linear regression problem where the target variable $y^{(r)} \in \mathbb{R}^n$ represents daily screen time in hours. The goal is to estimate a function that predicts continuous screen time values based on behavioral features.

2.3 Goals

To solve these problems, multiple machine learning models are applied including logistic regression, k-nearest neighbors, decision tree, and neural network for classification and linear regression for the regression task. These models allow for comparison of different approaches in capturing relationships within the data. The overall goal is to compute accurate predictions and evaluate model performance using different metrics. For classification, performance is assessed using accuracy, precision, recall, and F1 score along with confusion matrices and 5-fold cross-validation. For regression, performance is assessed using metrics such as mean squared error and R^2. The analysis tries to identify which behavioral features are most influential and to understand the strengths and limitations of each modeling approach.

Although decision trees were not covered in class, they were included in this project because of their ability to model complex nonlinear relationships. A decision tree works by repeatedly splitting the dataset into smaller groups based on feature values that best separate the target classes. These splits form a tree-like structure where each internal node represents a decision rule and each leaf node represents a final classification prediction.

3 Methodology

3.1 Preprocessing

Prior to model training, the dataset was preprocessed. Irrelevant identifier columns including transaction and user IDs were removed as they do not contribute to predictive performance. Missing values were handled by removing incomplete rows to maintain consistency across features.

3.2 One-hot encoding

One-hot encoding was applied to nominal variables such as gender and academic/work impact while ordinal variables such as stress level and addiction level were encoded using label encoding to preserve their ordering. The dataset was then separated into input features and target variables.

3.3 Standardization

The data was partitioned into training and testing sets using a 80/20 split to evaluate model performance on unseen data. Feature scaling was applied using standardization to ensure that all variables are on a comparable scale. Prior to model training, the distribution of the target variable was examined to assess class imbalance. The dataset exhibits a noticeable imbalance with approximately 79.45% of instances labeled as addicted and 20.55% as non-addicted. Such imbalance can bias models toward the majority class, leading to misleading accuracy and poor minority class detection.

To address this, class weighting was applied during model training. Specifically, weights are assigned inversely proportional to class frequencies computed as $w_c = \frac{n}{k \cdot n_c}$, where n is the total number of samples, k is the number of classes, and n_c is the number of samples in class c. This approach penalizes misclassification of the minority class more heavily and improves recall for underrepresented instances. Class weighting was applied to models that natively support it such as logistic regression and decision trees. In contrast, K-nearest neighbors and the neural network were trained on the original imbalanced dataset as handling imbalance in these models would require alternative approaches such as resampling or modifying the loss function, which were not implemented in this project.

4 Experiments, observations, discussion

4.1 Logistic Regression

Logistic regression was applied as a baseline model for the binary classification task of predicting smartphone addiction.
Hyperparameters:

Solver: lbfgs

Penalty: L2 regularization

Regularization strength (C): 1.0

Maximum iterations: 1000

Class weight: balanced

```
... Accuracy: 0.8302169035153328
                  precision    recall  f1-score   support

               0       0.55      0.84      0.66       265
               1       0.95      0.83      0.89      1072

        accuracy                           0.83      1337
       macro avg       0.75      0.83      0.77      1337
    weighted avg       0.87      0.83      0.84      1337

Training Time: 0.024363 seconds
Prediction Time: 0.002295 seconds
```

Figure 1: Statistics for Logistic Regression

The logistic regression model achieved an accuracy of 0.83 with strong performance on the majority class and improved recall for the minority class. This improvement in minority class recall can be attributed to the use of class weighting. However, this comes at the cost of lower precision (0.55) for the minority class indicating a tendency to over-predict non-addicted users. The model was computationally efficient with a training time of 0.024 seconds and a prediction time of 0.002 seconds

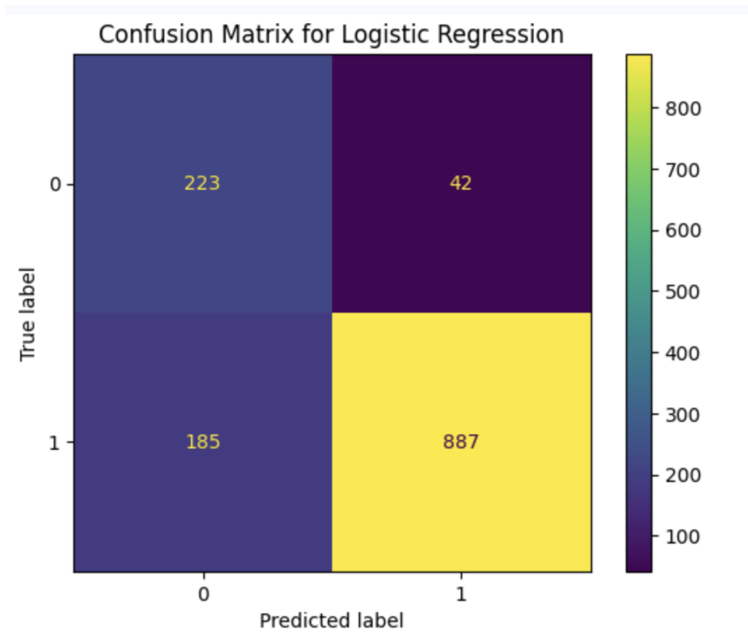

Figure 2: Confusion Matrix for Logistic Regression

The confusion matrix shows that the model correctly classified 223 non-addicted users and 887 addicted users indicating strong overall predictive performance. However, there are 42 false positives and 185 false negatives highlighting a tradeoff between precision and recall particularly for the minority class.

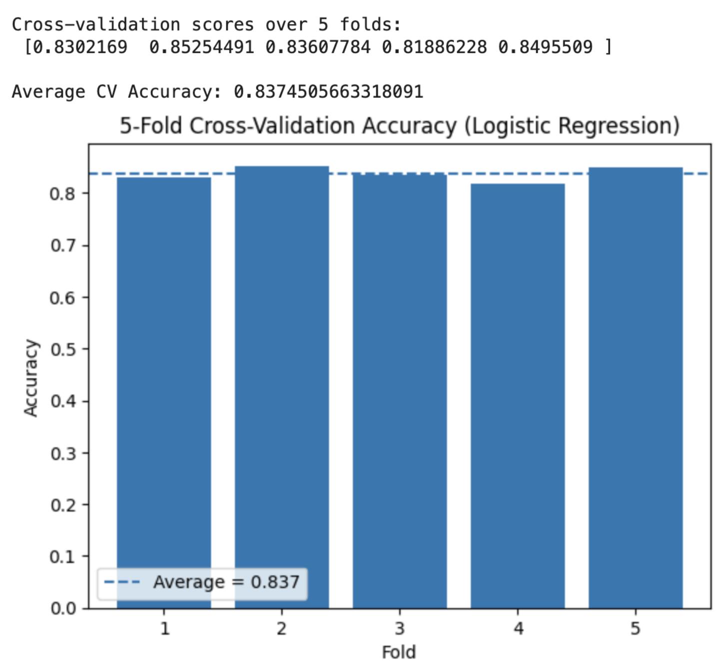

Figure 3: Cross-Validation for Logistic Regression

The 5-fold cross-validation results for logistic regression show accuracies ranging from approximately 0.82 to 0.85 with an average accuracy of 0.837. The relatively small variation across folds indicates stable and consistent model performance.

4.2 K-Nearest Neighbors

K-nearest neighbors was applied as an alternative classification model to compare performance against logistic regression in predicting smartphone addiction
Hyperparameters:

Number of neighbors: 5

Weights: distance

Distance metric: Euclidean

```
... Accuracy: 0.8825729244577412
                  precision    recall  f1-score   support

               0       0.76      0.59      0.67       265
               1       0.90      0.95      0.93      1072

        accuracy                           0.88      1337
       macro avg       0.83      0.77      0.80      1337
    weighted avg       0.88      0.88      0.88      1337

    Training Time: 0.014333 seconds
    Prediction Time: 0.189279 seconds
```

Figure 4: Statistics for K-Nearest Neighbors

The K-nearest neighbors model achieved an accuracy of 0.88, showing improved overall performance compared to logistic regression particularly on the majority class. However, unlike logistic regression, KNN does not incorporate class weighting and was trained on the original imbalanced dataset. This contributes to its lower recall for the minority class (0.59) indicating difficulty in correctly identifying non-addicted users. The model had a very fast training time of 0.014 seconds but a significantly higher prediction time of 0.189 seconds due to the need to compute distances to training samples.

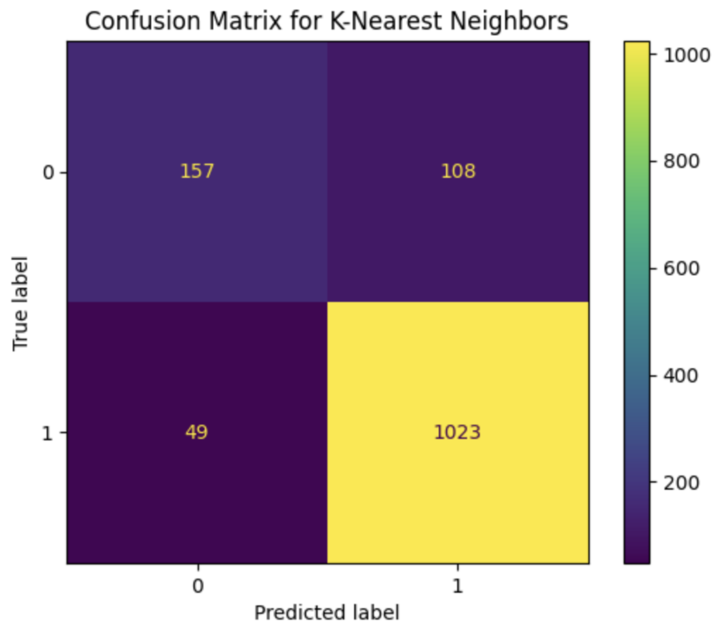

Figure 5: Confusion Matrix for K-Nearest Neighbors

The confusion matrix shows that the model correctly classified 157 non-addicted users and 1023 addicted users demonstrating strong overall predictive performance on the majority class. However, there are 108 false positives and 49 false negatives indicating that while the model is highly accurate overall, it struggles more with correctly identifying the minority class.

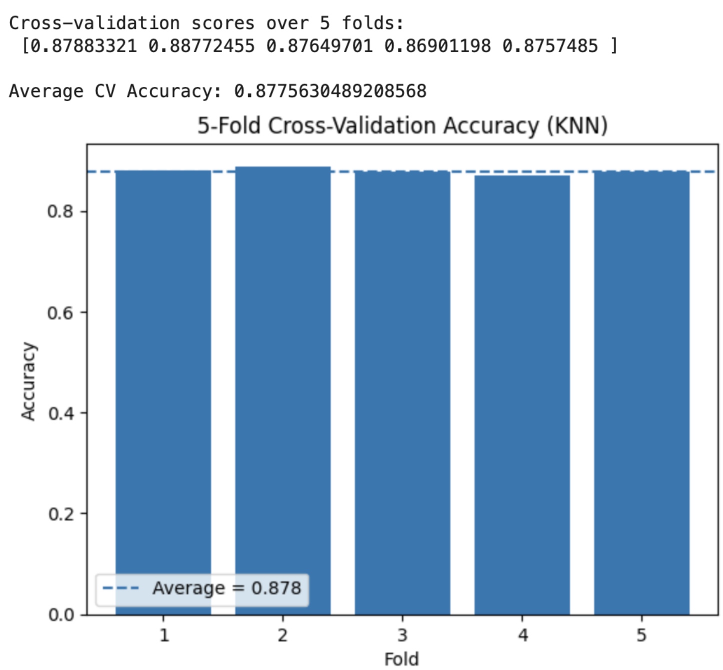

Figure 6: Cross-Validation for K-Nearest Neighbors

The KNN model achieved cross-validation accuracies between approximately 0.87 and 0.89 with an average accuracy of 0.88. The consistency across folds suggests reliable performance and overall accuracy is higher compared to logistic regression.

4.3 Decision Tree

A decision tree classifier was applied as a nonlinear model to capture complex relationships in the data for the binary classification task of predicting smartphone addiction.
Hyperparameters:

Criterion: gini

Max depth: None

Min samples split: 2

Random state: 42

Class weight: balanced

```
··· Accuracy: 0.9274495138369484
                  precision    recall  f1-score   support

               0       0.82      0.82      0.82       265
               1       0.96      0.95      0.95      1072

        accuracy                           0.93      1337
       macro avg       0.89      0.89      0.89      1337
    weighted avg       0.93      0.93      0.93      1337

    Training Time: 0.079069 seconds
    Prediction Time: 0.002436 seconds
```

Figure 7: Statistics for Decision Tree

The decision tree model achieved an accuracy of 0.93 demonstrating the highest overall performance among the models with strong precision and recall for both classes. This balanced performance is supported in part by the use of class weighting. The model maintained efficient computation with a training time of 0.079 seconds and a very fast prediction time of 0.002 seconds.

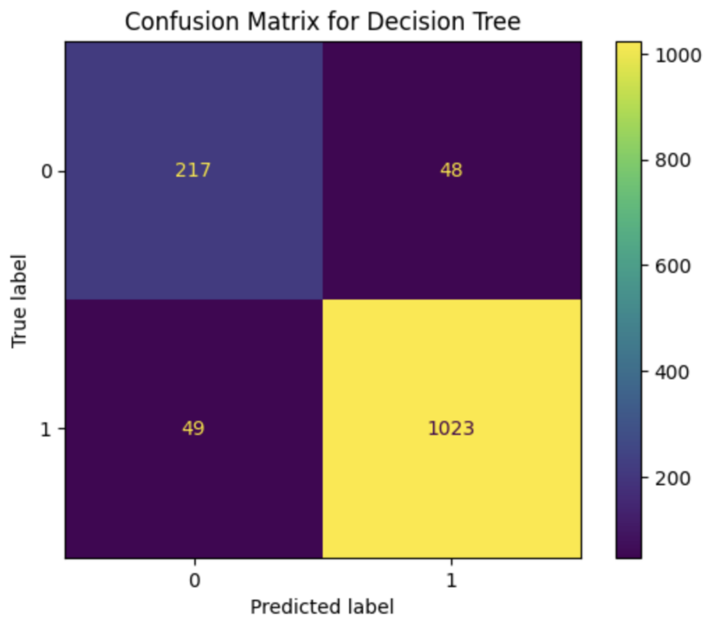

Figure 8: Confusion Matrix for Decision Tree

The confusion matrix shows that the model correctly classified 217 non-addicted users and 1023 addicted users indicating strong performance across both classes. There are 48 false positives and 49 false negatives showing a more balanced error distribution compared to the other models.

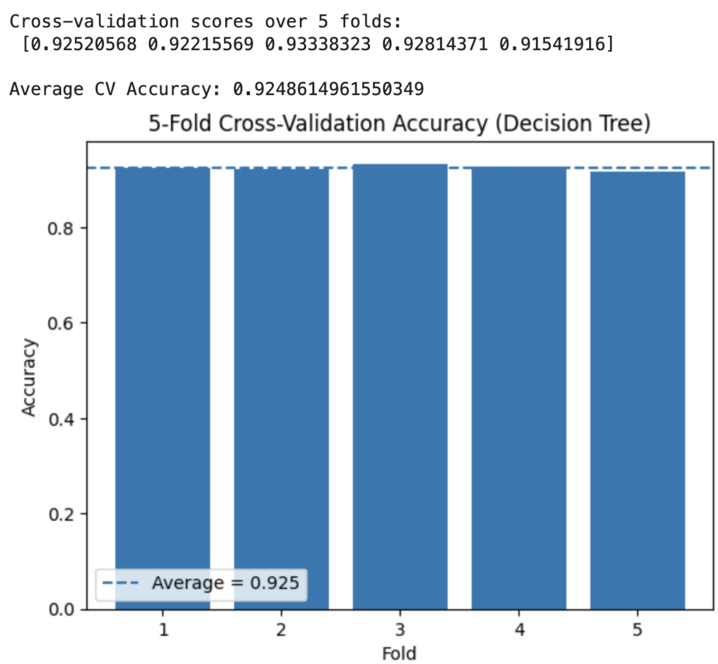

Figure 9: Cross-Validation for Decision Tree

The decision tree model produced high cross-validation accuracies ranging from approximately 0.91 to 0.93 with an average accuracy of 0.925. The minimal variation across folds indicates strong and stable generalization performance.

4.4 Neural Network

A neural network model was applied as a more complex nonlinear model to capture deeper patterns in the data for the binary classification task of predicting smartphone addiction.
Hyperparameters:

Hidden layers: (128, 64)

Activation: ReLU

Solver: Adam

Max iterations: 1000

Learning rate: 0.001

```
... Accuracy: 0.9237097980553478
                  precision    recall  f1-score   support

               0       0.79      0.83      0.81       265
               1       0.96      0.95      0.95      1072

        accuracy                           0.92      1337
       macro avg       0.88      0.89      0.88      1337
    weighted avg       0.93      0.92      0.92      1337

    Training Time: 16.588846 seconds
    Prediction Time: 0.004087 seconds
```

Figure 10: Statistics for Neural Network

The neural network model achieved an accuracy of 0.92 demonstrating strong overall performance with high precision and recall for the majority class. However, unlike logistic regression and decision trees, class imbalance handling was not applied as the model was trained on the original dataset without class weighting. This may contribute to slightly less balanced performance on the minority class. The model came at a higher computational cost with a training time of 16.59 seconds while prediction remained efficient at 0.004 seconds.

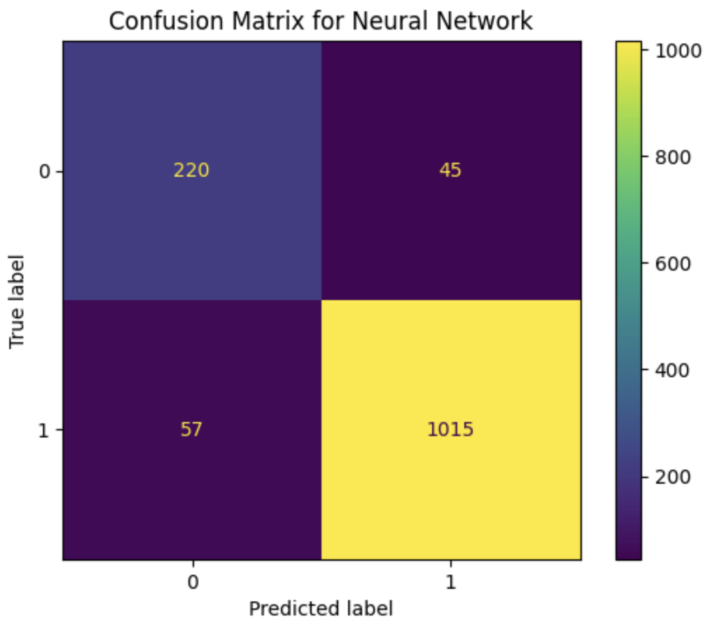

Figure 11: Confusion Matrix for Neural Network

The confusion matrix shows that the model correctly classified 220 non-addicted users and 1015 addicted users indicating strong performance across both classes. There are 45 false positives and 57 false negatives reflecting a relatively balanced error distribution while maintaining high overall accuracy.

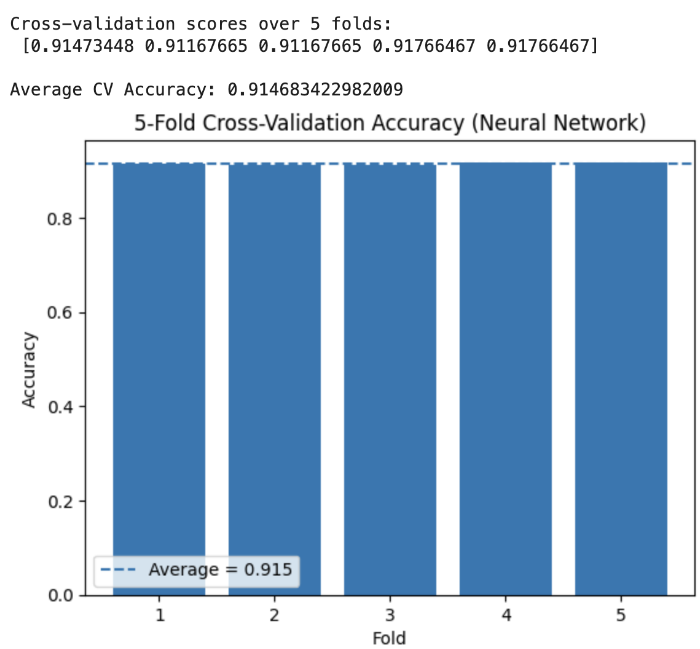

Figure 12: Cross-Validation for Neural Network

The neural network model achieved cross-validation accuracies between approximately 0.91 and 0.92 with an average accuracy of 0.915. The consistent results across folds demonstrate stable performance though slightly lower than the decision tree model.

4.5 Observations

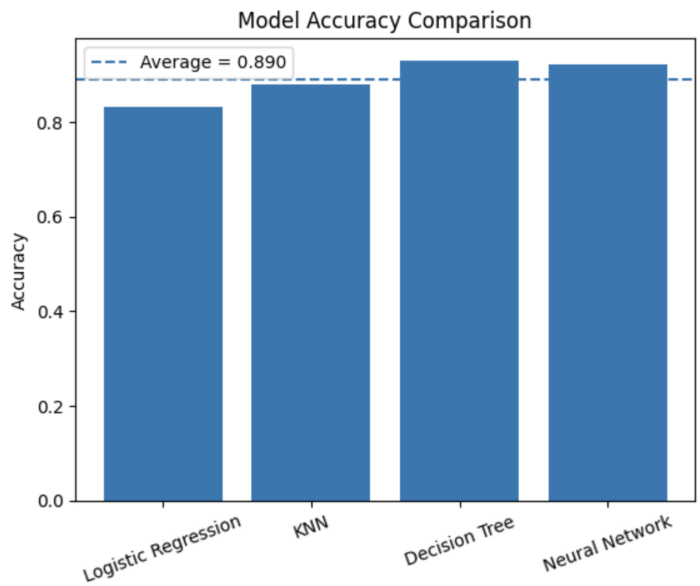

Figure 13: Accuracy Bar Chart

	Model	Accuracy	Precision (Class 0)	Recall (Class 0)	F1 (Class 0)	Precision (Class 1)	Recall (Class 1)	F1 (Class 1)	Training Time (s)	Prediction Time (s)
0	Logistic Regression	0.83	0.55	0.84	0.66	0.95	0.83	0.89	0.024	0.002
1	KNN	0.88	0.76	0.59	0.67	0.90	0.95	0.93	0.014	0.189
2	Decision Tree	0.93	0.82	0.82	0.82	0.96	0.95	0.95	0.079	0.002
3	Neural Network	0.92	0.79	0.83	0.81	0.96	0.95	0.95	16.589	0.004

Figure 14: Statistics for all Four Classification Models

The performance of the four classification models shows clear differences in both accuracy and class-level metrics. The decision tree achieved the highest accuracy (0.93), closely followed by the neural network (0.92), while K-nearest neighbors achieved moderate performance (0.88), and logistic regression produced the lowest accuracy (0.83). This suggests that nonlinear models such as decision trees and neural networks are better suited for capturing the complex relationships present in the dataset compared to linear models.

From a class-level perspective, all models performed significantly better on the majority class (addicted users) than the minority class. Logistic regression showed high recall for the minority class (0.84) but low precision (0.55) indicating a tendency to over-predict that class, which is consistent with the use of class weighting. In contrast, KNN improved precision for the minority class (0.76) but at the cost of lower recall (0.59), likely due to the absence of imbalance handling. Both the decision tree and neural network achieved a more balanced performance with precision and recall values around 0.80+ for the minority class with the decision tree benefiting from class weighting while the neural network maintained strong performance despite being trained on the original imbalanced dataset.

In terms of computational efficiency, logistic regression and KNN had very fast training times while the neural network required significantly more time (16.59 seconds) reflecting its higher computational complexity. However, prediction times remained relatively low for all models except KNN which showed slower prediction due to distance calculations. Overall, the decision tree provided the best balance between accuracy, class performance, and computational efficiency making it the most effective model for this classification task.

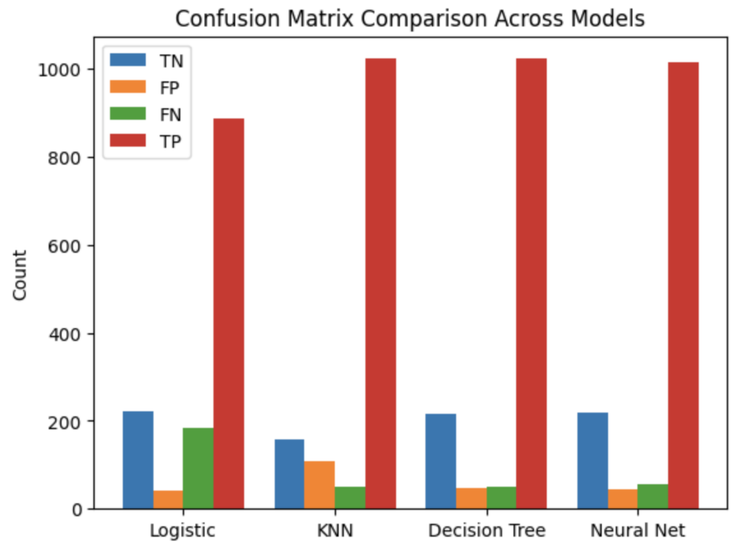

Figure 15: Confusion Matrix Values Bar Chart

	Model	True Negative (TN)	False Positive (FP)	False Negative (FN)	True Positive (TP)
0	Logistic Regression	223	42	185	887
1	KNN	157	108	49	1023
2	Decision Tree	217	48	49	1023
3	Neural Network	220	45	57	1015

Figure 16: Statistics for Confusion Matrix Values

The confusion matrix results provide a detailed comparison of how each model handles correct and incorrect classifications. The decision tree and neural network models demonstrate the strongest overall performance with high true positive counts (1023 and 1015) and relatively low false positive and false negative values indicating balanced and accurate predictions across both classes. Logistic regression, while achieving a reasonable number of true negatives (223), suffers from a high number of false negatives (185) suggesting difficulty in correctly identifying addicted users. In contrast, K-nearest neighbors achieves a high number of true positives (1023) but produces the largest number of false positives (108) indicating a tendency to incorrectly classify non-addicted users as addicted. Overall, these results highlight that nonlinear models, particularly the decision tree, are more effective at handling both classes while minimizing classification errors.

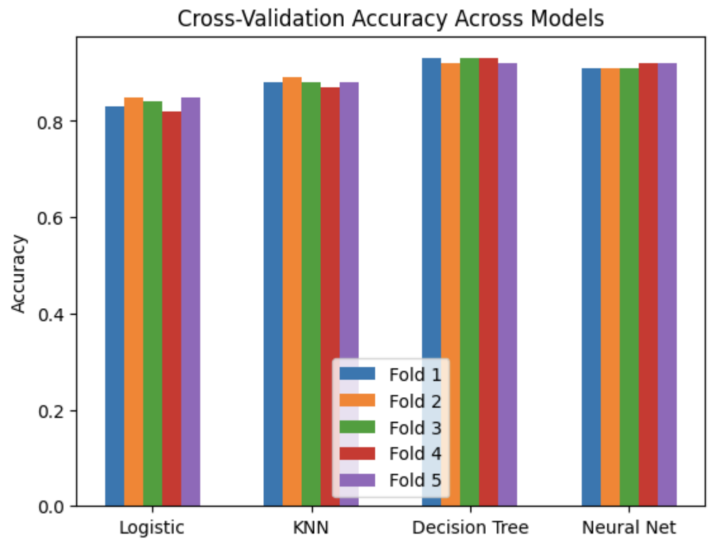

Figure 17: Cross-Validation Accuracy Bar Chart

	Model	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Average Accuracy
0	Logistic Regression	0.83	0.85	0.84	0.82	0.85	0.838
1	KNN	0.88	0.89	0.88	0.87	0.88	0.880
2	Decision Tree	0.93	0.92	0.93	0.93	0.92	0.926
3	Neural Network	0.91	0.91	0.91	0.92	0.92	0.914

Figure 18: Cross-Validation Accuracy Values

The cross-validation results demonstrate clear differences in model performance across the five folds. The decision tree achieved the highest average accuracy (0.926) followed closely by the neural network (0.914) indicating that these models generalize well to unseen data. K-nearest neighbors showed moderate performance with an average accuracy of 0.880 while logistic regression had the lowest average (0.838) suggesting linear models may be less effective for this dataset. Across all models, the variation between folds is relatively small indicating stable and consistent performance.

4.6 Discussion

Beyond performance metrics, this analysis seeks to determine which behavioral factors are most influential in predicting smartphone addiction. Identifying key features allows for a better understanding of the relationships driving model predictions. Feature importance was therefore analyzed using the decision tree model as it provided the highest accuracy and offers clear interpretability.

	Feature	Importance
1	daily_screen_time_hours	0.493339
2	social_media_hours	0.397188
4	work_study_hours	0.019468
6	notifications_per_day	0.017606
5	sleep_hours	0.016799
8	weekend_screen_time	0.016448
7	app_opens_per_day	0.013099
3	gaming_hours	0.011582
0	age	0.005910
11	gender_Male	0.003625
12	gender_Other	0.001524
9	stress_level	0.001169
13	academic_work_impact_No	0.001123
14	academic_work_impact_Yes	0.000746
10	gender_Female	0.000374

Figure 19: Top 10 Feature Importance Percentage

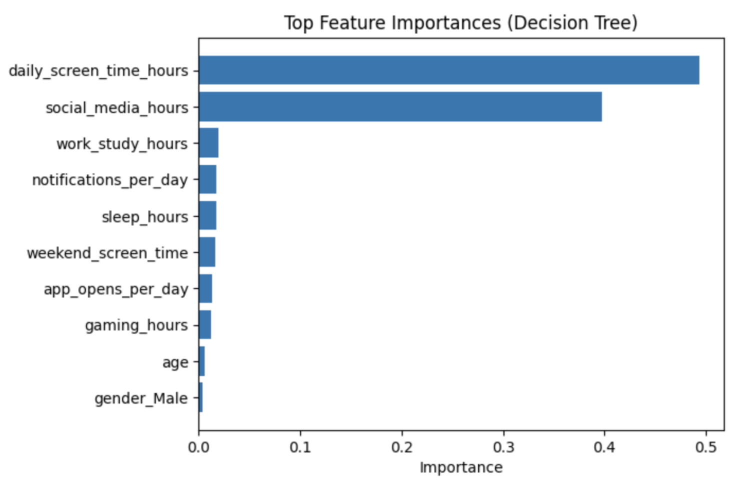

Figure 20: Top 10 Feature Importance Bar Chart

The feature importance results from the decision tree model indicate that daily screen time (0.493) and social media usage (0.397) are by far the most influential predictors of smartphone addiction accounting for the majority of the model's decision-making. All other features, including work hours, notifications per day, and sleep hours, have significantly smaller importance values indicating a relatively minor contribution to predictions. Demographic and categorical variables such as age, gender, and stress level show minimal influence suggesting that behavioral patterns are much more important than personal characteristics.

Beyond classification, a secondary task was formulated as a linear regression problem to predict daily screen time in hours. This task complements the classification analysis by modeling the intensity of smartphone usage and uncovering patterns in continuous behavior.

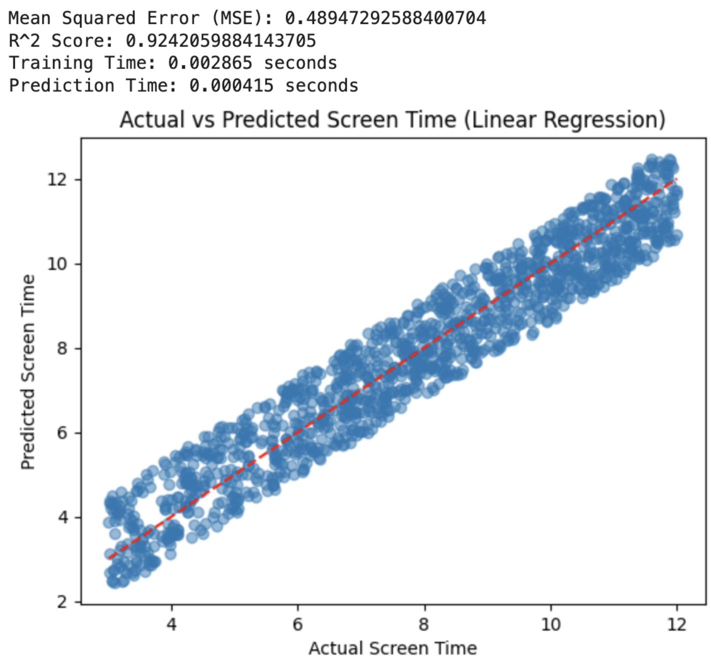

Figure 21: Prediction of Screen Time using Linear Regression

The linear regression model achieved a mean squared error of approximately 0.49 and an R^2 score of 0.924 indicating that over 92% of the variance in daily screen time is explained by the selected features. This reflects strong predictive performance with relatively low error given the continuous nature of the target variable. The scatter plot of actual versus predicted values shows a tight clustering around the diagonal reference line, further confirming that the model produces accurate and consistent predictions.

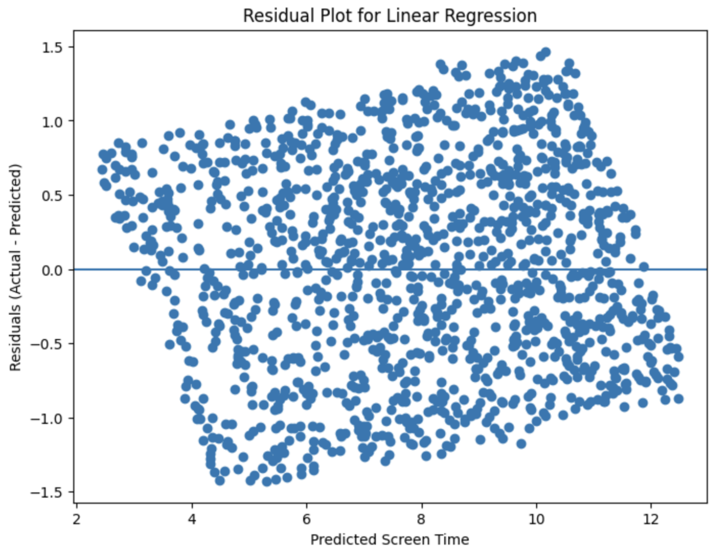

Figure 22: Prediction Screen Time Residual Plot

A residual plot was analyzed to evaluate the distribution of prediction errors in the linear regression model. The residuals are generally centered around zero indicating that the model does not exhibit strong systematic bias. However, there is a slight increase in the spread of residuals as predicted screen time increases. This indicates that the model is slightly less accurate for higher screen time values, although overall performance remains strong.

4.7 Limitations of the Experiment

While the results of this project demonstrate strong predictive performance, several limitations should be acknowledged:

1. The dataset consists of 7500 observations, which while sufficient for training machine learning models, may still limit generalizability of the results. A larger and more diverse dataset could provide a more comprehensive representation of smartphone usage behaviors across different populations.

2. The dataset exhibits a significant class imbalance with approximately 79% of instances labeled as addicted and only 21% as non-addicted. This imbalance can bias models towards the majority class and influence evaluation metrics. Although class weighting was applied to certain models, imbalance handling was not implemented consistently across all models which could affect the fairness of model comparisons. Furthermore, class imbalance is a common challenge in real-world behavioral datasets where the prevalence of high smartphone usage can naturally lead to a skewed distribution.

3. Additionally, the choice of imbalance handling method presents its own limitations. While class weighting is computationally efficient and preserves the original data distribution, alternative approaches such as oversampling (e.g., SMOTE) or undersampling could potentially improve minority class performance. However, these methods introduce tradeoffs including the risk of overfitting or loss of important information.

4. Another limitation lies in the feature set used for analysis. Although the dataset includes key behavioral variables such as screen time and app usage, it contains a relatively limited

number of features and may not capture all relevant factors influencing smartphone addiction. Additional variables such as psychological or environmental data (e.g., measures of anxiety or well-being, social environment or peer influence) could provide deeper insight into user behavior.

5. Finally, while multiple machine learning models were explored, the project primarily focused on standard algorithms. More advanced or nonlinear approaches, such as support vector machines, ensemble methods (e.g., random forests or gradient boosting), or deeper neural network architectures may further improve predictive performance and better capture complex relationships within the data.

5 Conclusion

In this project, machine learning models were applied to analyze and predict smartphone addiction using real-world usage data. The results demonstrate that nonlinear models, particularly the decision tree and neural network outperform linear approaches in capturing complex behavioral patterns. Among the models, the decision tree provided the best overall balance between accuracy, class-level performance, and computational efficiency. Feature importance analysis revealed that behavioral variables such as screen time and social media usage are the most influential predictors while demographic and contextual factors play a comparatively smaller role. The regression analysis further supported these findings by accurately predicting daily screen time indicating strong relationships between usage behavior and model inputs. Overall, this project highlights the effectiveness of machine learning in understanding smartphone addiction while emphasizing the importance of behavioral patterns in predicting user outcomes.

6 Reference

Smartphone Usage and Addiction Analysis Dataset. *Kaggle*. https://www.kaggle.com/datasets/zahranusratt/smartphone-usage-and-addiction-analysis-dataset?resource=download